

ASSIGNMENT

- 1 (ASUMO/1988) There are two piles of pebbles, pile (A) and pile (B). When 100 pebbles are moved from (A) to (B), then the number of pebbles in (B) is double of that in (A). However, if some are moved from (B) to (A), then the number of pebbles in (A) is five times more than that in (B). What is the minimum possible number of pebbles in (A), and find the number of pebbles in (B) in that case.

Solution:

Let x and y be the numbers of pebbles in the piles (A) and (B) respectively. When z pebbles are moved from (B) to (A), then the given conditions in question gives

$$2(x - 100) = y + 100 \quad \text{gives } y = 2x - 300$$

$$x + z = 6(y - z)$$

So from it follows that $11x - 7z = 1800$

$$\Rightarrow 4x + 7(x - z) = 1800$$

Both sides taking modulo 4 yields $4 \mid (x - z)$, so $x - z = 4t, t \in \mathbb{Z}$.

Implies $4x + 28t = 1800$ or $x + 7t = 450$.

Thus, the general solution for $(x; y; z)$ is

$$x = 450 - 7t, y = 2(450 - 7t) - 300 = 600 - 14t, z = (450 - 7t) - 4t = 450 - 11t.$$

From $x, y, z \geq 0$, it is obtained that $t \leq \frac{450}{11} < 41$, so $t \leq 40$. When t takes its maximum possible value

then x is its minimum, so $x_{\min} = 450 - 280 = 170$

In that case, $y = 600 - 560 = 40, z = 450 - 440 = 10$, so there are 40 pebbles in (B)

- 2 (CHINA/1990) An integer solution of the equation $1990x - 1989y = 1991$ is

Option:

(a) $x = 12785; y = 12768$; (b) $x = 12785; y = 12770$; (c) $x = 11936; y = 11941$;

(d) $x = 13827; y = 12623$.

Answer: (c)

Solution:

The given equation implies $y \equiv 1 \pmod{10}$, only C is possible. By checking (C) is a solution. Thus the answer is (C).

- 3 (SSSMO(J)/2002) Two positive integers A and B satisfy $\frac{A}{11} + \frac{B}{3} = \frac{17}{33}$ Find the value of

$$A^2 + B^2$$

Solution:

The given equality yields $\frac{3A + 11B}{33} = \frac{17}{33}$, so It is easy to find the special solution (2,1) for(A,B). Since

the general solution is $A = 2 + 11t, B = 1 - 3t, t \in \mathbb{Z}$,

$A \geq B$ and $B \geq 1$ implies that $t = 0$ is the unique permitted values of t , so (2,1) is the unique desired solution for (A,B). Thus $A^2 + B^2 = 5$.

- 4 A dragonfly has six feet and a spider has 8 feet. Given that a certain group of dragonflies and spiders have in total 46 feet, find the number of dragonflies and the number of spiders.

Solution:

Let x and y be the numbers of dragonflies and spiders respectively. Then

$$6x + 8y = 46 \text{ or } 3x + 4y = 23$$

It is clear that $x \neq 0$ and $y \neq 0$ since 46 is neither divisible by 6 nor by 8.

$$4y < 23 \text{ implies } y \leq 5,$$

so corresponding to $y = 1, 2, 3, 4, 5, 6$ the equation gives

$$x = \frac{23 - 4y}{3}$$

Can be positive integer at $y = 2$ or $y = 5$ only $x = 5$ if $y = 2$ and $x = 1$ at $y = 5$.

So the answer is that 5 dragonflies and 2 spiders or one dragonfly and 5 spiders

- 5 If a four digit number and the sum of its all digits have a sum 2006, find the four digit number.

Solution:

Let the four digit number be $\overline{abcd} = 1000a + 100b + 10c + d$

$$\text{Then } 1001a + 101b + 11c + 2d = 2006$$

Which implies that $a = 1$ or $a = 2$

For $a = 1$,

$$\Rightarrow 101b + 11c + 2d = 1005.$$

$$\text{Since } 101 \times 8 + 11 \times 9 + 2 \times 9 = 925 < 1005,$$

Therefore $b = 9$, so $11c + 2d = 96$. $11 \times 7 + 2 \times 9 = 95 < 96$ and $c < 9$ then gives $c = 8$,

$$\text{So } 2d = 8 \text{ ie. } d = 4$$

Thus, the four digit number is 1984.

For $a = 2$

$$101b + 11c + 2d = 4.$$

$$b = c = 0 \text{ \& } d = 2$$

Thus, the four digit number is 2002.

- 6 (Ancient Question) In an ancient chicken market, each rooster is sold for 5 coins, each hen for 3 coins and each chick for $\frac{1}{3}$ coins. Someone has 100 coins to buy 100 chickens, how many roosters, hens and chicks can a man purchase out of a total cost of 100 coins?

Solution:

Let the numbers of roosters, hens and chicks that the buyer bought be x, y, z respectively then

$$5x + 3y + \frac{1}{3}z = 100$$

$$x + y + z = 100$$

Or

$$15x + 9y + z = 300, \quad (i)$$

$$x + y + z = 100 \quad (ii)$$

(i) – (ii) yields

$$14x + 8y = 200 \text{ or, equivalently, } 7x + 4y = 100 \quad (v)$$

It has special solution $x_0 = -100, y_0 = 200$,

So the general solution for (x, y) is $x = -100 + 4t$,

$y = 200 - 7t$, where t is any integer.

The equation (v) indicates that $4 \mid x$, so x maybe 0, 4, 8, 12, 16, 20 only.

Correspondingly, $t = 25, 26, 27, 28, 29, 30$ only. Since $y \geq 0$ implies

$t \leq 200/7 < 29$, so $t = 25, 26, 27, 28$, only i.e. $y = 25, 18, 11, 4$.

Thus $x + y$ is 25, 22, 19, 16, respectively so $z = 75, 78, 81, 84$ correspondingly.

Thus, the solutions for (x, y, z) are (0, 25, 75), (4, 18, 78), (8, 11, 81), (12, 4, 84)

- 7 Solve the equation $3x + 5y = 7$ in integers.

Solution:

$$3(-1) + 5(2) = 7.$$

$$x = -1 + 5k$$

$$y = 2 - 3k$$

Where k is an arbitrary integer.

- 8 Solve the equation $1990x - 173y = 11$ in integers.

Solution:

The coefficients in the equation are large enough to make it difficult to find a particular solution. However, it is not hard to see that the numbers 1990 and 173 are relatively prime, and this helps. The greatest common divisor (gcd) of these numbers can be represented as $1990m - 173n$ for some integers m and n

Euclid's algorithm gives us $m = 2, n = 23$.

Hence $x_0 = 2 \times 11 = 22, y_0 = 23 \times 11 = 253$ is a solution to the equation $1990x - 173y = 11$

$$x = x_0 + 173k = 22 + 173k,$$

$$y = y_0 + 1990k = 253 + 1990k, \quad \text{where } k \text{ is any integer}$$

- 9 Find all integer roots of the equations $21x + 48y = 6$.

Solution:

$$21x + 48y = 6 \Rightarrow 7x + 16y = 2$$

The solutions are of the form $\{x = 16k - 2, y = -7k + 1\}$, where k takes on all integer values

- 10 Find all integer roots of the equation $x^2 + y^2 = 4z - 1$

Solution:

It is clear that one cannot transform the equation to a more tractable type; it is also impossible to analyze all the appropriate triples of integers. This new representative of our "Diophantine zoo" is remarkable in that it has no integer solutions. Indeed, Which remainders can perfect squares give when divided by 4? The choice of modulo 4 was determined by the form of the given equation). The only possible remainders are 0 and 1. Since the sum of two such remainders cannot give remainder -1 , we have no solutions for this equation

- 11 Solve for integer values of the variables: $x^3 + 21y^2 + 5 = 0$

Solution:

Because x^3 can be congruent modulo 7 only to 0, 1, -1 and $x^3 + 21y^2 + 5$ must be congruent modulo 7 to 5, 6 or 4 and, therefore, cannot be zero

- 12 Solve for integer values of the variables: $15x^2 - 7y^2 = 9$

Solution:

$$15x^2 - 7y^2 \equiv 9 \pmod{5}$$

$$\Rightarrow 0 - 2y^2 \equiv 4 \pmod{5}$$

$$\Rightarrow 3y^2 \equiv 4 \pmod{5}$$

$$\text{But } 3y^2 \equiv 0, 2, 3 \pmod{5}$$

No solution possible.

- 13 Solve for integer values of the variables: $x^2 - y^2 = 1988$

Solution:

The answers are $x = \pm 498$, $y = \pm 496$ and $x = \pm 78$, $y = \pm 64$ (the signs can be chosen independently).

Rewrite the equation as follows : $(x-y)(x+y) = 2 \times 2 \times 7 \times 71$ (the number 71 is prime). We can temporarily assume that x and y are positive (later we can supply these numbers with arbitrary signs). We have only two representations of the number 1988 as the product of two positive integers of equal parity: $1988 = 2 \times 994$ and $1988 = 14 \times 142$.

Setting the factors $x - y$, $x + y$ equal to these completes the solution

- 14** (CHNMO(P)/2002) A positive integer is called a “good number” if it is equal to four times of the sum of its digits. Find the sum of all good numbers.

Solution:

Let $\overline{ab} = 10a + b$ be a two digit good number, then $10a + b = 4(a + b)$ implies $2a = b$, so there are four good numbers 12, 24, 36, 48, and their sum is 120.

Three digit good number $\overline{abc}$ satisfies the equation $100a + 10b + c = 4(a + b + c)$, i.e. $96a + 6b - 3c = 0$. Since $96a + 6b - 3c \geq 96 + 0 - 27 > 0$ always so no solution for $(a; b; c)$, i.e. no three digit good number exists.

Since a number with n ($n \geq 4$) digits must be not less than 10^{n-1} and the 4 times of the sum of its digits is not greater than $36n$. For $n \geq 4$,

$$10^{n-1} - 36n > 36(10^{n-3} - n) > 0,$$

So no n digit good number exists if $n \geq 4$

Thus, the sum of all good numbers is 120.

- 15** (MOSCOW/1940) Find all three-digit numbers such that each is equal to the sum of the factorials of its own digits.

Solution:

Let $\overline{abc} = 100a + 10b + c$ be desired three digit number $7! = 5040$ indicates that $a, b, c \leq 6$, and further,

if one of a, b, c is 6 then $\overline{abc} > 6! = 720 \Rightarrow$ one a, b, c is greater than 6,

So $a, b, c \leq 5$. Since $\overline{555} \neq 5! + 5! + 5!$. So a, b, c cannot be all 5

On the other hand, $4! + 4! + 4! = 72$ which is not a three digit number,

So at least one of a, b, c is 5.

$$\overline{abc} < 5! + 5! + 5! = 360 \text{ implies that } a \leq 3$$

When $a = 1$, then $145 = 1! + 4! + 5!$, so 145 is a desired number

When $a = 2$, then b, c must be 5. But $255 \neq 2! + 5! + 5!$, so no solution

When $a = 3$, then b, c must be 5. But $355 \neq 3! + 5! + 5!$ so no solution.

Thus, 145 is the unique solution.